L-theanine, a natural constituent in tea, and its effect on mental state.

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Tea is the most widely consumed beverage in the world after water. Tea is known to be a rich source of flavonoid antioxidants. However tea also contains a unique amino acid, L-theanine that may modulate aspects of brain function in humans. Evidence from human electroencephalograph (EEG) studies show that it has a direct effect on the brain (Juneja et al. Trends in Food Science & Tech 1999;10;199-204). L-theanine significantly increases activity in the alpha frequency band which indicates that it relaxes the mind without inducing drowsiness. However, this effect has only been established at higher doses than that typically found in a cup of black tea (approximately 20mg). The aim of the current research was to establish this effect at more realistic dietary levels. EEG was measured in healthy, young participants at baseline and 45, 60, 75, 90 and 105 minutes after ingestion of 50mg L-theanine (n=16) or placebo (n=19). Participants were resting with their eyes closed during EEG recording. There was a greater increase in alpha activity across time in the L-theanine condition (relative to placebo (p+0.05). A second study replicated this effect in participants engaged in passive activity. These data indicate that L-theanine, at realistic dietary levels, has a significant effect on the general state of mental alertness or arousal. Furthermore, alpha activity is known to play an important role in critical aspects of attention, and further research is therefore focussed on understanding the effect of L-theanine on attentional processes.